Kapasa Makasa University

Information Communication Technology Department
(Cyber Security)

Wireless Security

(CYS331)

Unit 5

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WIRELESS SECURITY THREATS

Threat

- **A threat** is any potential danger that can exploit a vulnerability in a system or network to cause harm, damage, or unauthorized access to data, resources, or services.
- In other words, it is anything that can **compromise the confidentiality, integrity, or availability** of information or systems.

Security Breach

- A **security breach** is an incident where unauthorized individuals gain access to a system, network, or data, violating the system's security policies. This access can lead to the theft, exposure, alteration, or destruction of sensitive information or disruption of services.
- In short, it's when someone breaks into a protected system or data without permission, causing a compromise in confidentiality, integrity, or availability.

Difference Between Threat and Breach

Aspect	Threat	Breach
Timing	Before the attack	After the attack
Action	Potential attack	Successful attack
Impact	No damage yet	Actual damage/loss
Examples	Malware, phishing attempts, unpatched software	Data theft, system compromise, information leak

Security Threats

- Eavesdropping
- Communication jamming
- RF Interference
- Spoofing
- SQL Injection (Software Layer)

Eavesdropping

- **Eavesdropping** is a type of passive attack where an unauthorized person secretly listens to or captures private communications over a network.
- Eavesdropping is especially dangerous because wireless signals travel through the air and can be intercepted without physical access.

How It Works:

- The attacker uses tools (like packet sniffers) to monitor network traffic.
- If the communication is not encrypted, sensitive data such as passwords, emails, or credit card numbers can be stolen.
- **Example:**
- An attacker sits in a coffee shop with a laptop and uses Wireshark to capture data from users connected to the public Wi-Fi.

Eavesdropping

Types of Eavesdropping:

Passive Eavesdropping – Just listens to data without altering it.

Active Eavesdropping – May inject queries or interfere to gather more data.

Prevention:

Use encryption (WPA3 for Wi-Fi, SSL/TLS for web).

Use VPNs on public networks.

Avoid using public Wi-Fi for sensitive tasks.

Communication Jamming

Communication jamming refers to a **deliberate attempt to disrupt, block, or interfere with wireless communications** by transmitting signals that interfere with the normal operation of communication systems. It is a form of Denial of Service (DoS) attack specifically aimed at wireless communication channels

In wireless communication (e.g., Wi-Fi, mobile networks, GPS, military radios), devices transmit signals using radio frequencies. A jammer floods the frequency with **noise** or **fake signals**, preventing legitimate devices from communicating effectively.

Types of Jamming

1. **Spot Jamming**
    1. Targets a specific frequency with a high-power signal.
    2. Used to block a particular channel (e.g., a Wi-Fi channel).
2. **Sweep Jamming**
    1. Continuously sweeps across a range of frequencies.
    2. Briefly disrupts multiple frequencies over time.
3. **Barrage Jamming**
    1. Simultaneously jams a wide range of frequencies.
    2. Consumes a lot of power but is effective against broad-spectrum communication.
4. **Deceptive Jamming**
    1. Sends false information or signals to confuse the receiving system.
    2. Example: Fake GPS signals (spoofing) to mislead navigation systems.

Effects and Measures

Effects and Risks

- **Loss of communication** between security forces or emergency services.
- **Interruption of Wi-Fi, mobile, GPS, or drone systems.**
- **Safety hazards** in aviation, military, and health systems.
- May cause **loss of service** in public or commercial networks.

Countermeasures

- **Frequency hopping**: Changes communication frequency rapidly to avoid jammers.
- **Spread spectrum**: Spreads signal across a wide frequency band.
- **Shielding and filtering**: Hardware solutions to reduce interference.
- **Signal detection**: Tools to detect and locate jammers (e.g., spectrum analyzers).

RF Interference (Radio Frequency Interference)

RF Interference is the unwanted disturbance caused by external radio frequency signals that disrupt the normal operation of electronic devices and communication systems. It occurs when two or more devices use the same or overlapping frequencies, leading to signal degradation, noise, or complete communication failure

RF interference can be **intentional** (like jamming) or **unintentional** (accidental).

Sources of RF Interference

Common sources include:

Source	Description
Electronic Devices	Microwaves, Bluetooth devices, cordless phones, baby monitors.
Power Lines	Arcing or faulty lines emitting RF noise.
Other Wireless Networks	Overlapping Wi-Fi signals in crowded areas (e.g., apartments).
Industrial Equipment	Motors, generators, or welding equipment.
Weather or Natural Sources	Lightning and solar flares.

Types and Effects of RF

Types of RF Interference

1. **Co-channel interference:**
    1. Occurs when two transmitters use the **same frequency channel**.
    2. Common in cellular networks.
2. **Adjacent-channel interference:**
    1. Happens when signals from **nearby channels** overlap.
3. **Intermodulation interference:**
    1. Caused when multiple RF signals mix together and produce **unwanted frequencies**.
4. **Harmonic interference:**
    1. Comes from signals that are **multiples of a base frequency**, unintentionally picked up by other devices.

Effects of RF Interference

- **Dropped calls** in mobile networks.
- **Poor Wi-Fi performance** or low data speeds.
- **Disrupted broadcast signals** (TV, radio).
- **Errors in sensitive instruments** (e.g., medical or aviation equipment).
- **Interruption of IoT or industrial wireless devices.**

Prevention & Mitigation

Technique	How it Helps
Shielding	Physical barriers (like metal enclosures) to block RF.
Filtering	Uses filters to remove unwanted frequencies.
Proper Antenna Placement	Avoids overlap and interference zones.
Channel Planning	Assigns non-overlapping channels (especially in Wi-Fi).
Spread Spectrum Techniques	Spreads signal across wider frequency to reduce vulnerability.

SQL Injection (Attacks the Software Layer)

SQL Injection (SQLi) is a code injection attack where a malicious user inserts or "injects" harmful SQL code into input fields or parameters of an application. This allows attackers to interfere with the queries an application makes to its database.

- Unauthorized access to sensitive data (e.g., usernames, passwords, financial records).
- Bypass login systems (e.g., logging in without a valid password).
- Modify or delete data in the database.
- Execute administrative operations (like dropping tables).
- Control the entire database server in severe cases.

Example of an Injection

- Suppose a login form sends the following SQL query

SELECT * FROM users WHERE username = 'admin' AND password = '1234’;

An attacker enters this instead:

Username: admin

Password: ' OR '1'=‘1

Now the query becomes:

SELECT * FROM users WHERE username = 'admin' AND password = '' OR '1'='1';

How to Prevent SQL Injection

- Use Prepared Statements / Parameterized Queries
- Sanitize and validate all user inputs
- Use ORM frameworks (e.g., SQLAlchemy, Hibernate)
- Apply least privilege principle in database access
- Regularly update and patch software

Denial of Service (Dos)

A Denial of Service (DoS) attack is a malicious attempt to make a computer, server, or network resource unavailable to its intended users by overwhelming it with a flood of fake traffic or requests.

Imagine a website can handle 100 users at a time. If an attacker sends thousands of fake requests every second, the server becomes overloaded and can't serve real users — it crashes, slows down, or becomes completely unreachable.

Types of DoS Attacks:

1. **Volume-Based Attacks**
    1. Flood the target with huge amounts of traffic (e.g., **UDP Flood**, **ICMP Flood**).
    2. Goal: Consume **bandwidth**.
2. **Protocol Attacks**
    1. Exploit weaknesses in network protocols (e.g. **Ping of Death**).
    2. Goal: Exhaust **server resources** like CPU or memory.
3. **Application Layer Attacks**
    1. Target specific apps or services (e.g., **HTTP Flood**).
    2. Goal: Exhaust the **application's resources**.

Spoofing

Spoofing is a cyberattack where an attacker pretends to be someone else by faking data to trick a system or user into trusting them.

This can be done on:

- IP Address (IP Spoofing): Faking the source IP to hide identity or impersonate a trusted device.
- Email Address (Email Spoofing): Sending emails that appear to come from a legitimate sender.
- MAC Address (MAC Spoofing): Changing the hardware address to bypass network controls.
- Caller ID (Caller ID Spoofing): Faking phone numbers to impersonate someone else.
- Website/URL (Website Spoofing): Creating fake sites resembling legitimate ones to steal information.

Why spoof and Mitigation

Reasons for spoofing

- To bypass security controls.
- To launch man-in-the-middle (MITM) attacks.
- To steal data or credentials.
- To spread malware.
- To hide their true identity.

How to Protect Against Spoofing

- Use **encryption** (e.g., HTTPS, VPN).
- Implement **email authentication** (**SPF** (Sender Policy Framework) , **DKIM** (DomainKeys Identified Mail), **DMARC** (Domain-based Message Authentication, Reporting and Conformance)).
- Employ **network security measures** like firewalls and intrusion detection.
- Educate users about suspicious emails and links.

END